Homework #9. Due on Thursday, Nov 4th, 11:59pm on Canvas

Reading for this homework: 5.1 and parts of 5.2

Plan for next week (Nov 2,4): 5.2 (Diagonalizability) continued, 7.3 (Minimal Polynomials) and 5.4 (Invariant subspaces and the Cayley-Hamilton Theorem)

Problems:

Problem 1: Let F be a field. Recall that the trace of a square matrix over F denoted by tr is the sum of its diagonal entries. Also recall from HW#1.2 that $\mathrm{tr}(AB) = \mathrm{tr}(BA)$ for any $A, B \in \mathrm{Mat}_n(F)$.

(a) Prove that if $A, C \in \mathrm{Mat}_n(F)$ are similar (that is, $C = Q^{-1}AQ$ for some invertible $Q \in \mathrm{Mat}_n(F)$), then $\mathrm{tr}(A) = \mathrm{tr}(C)$.

(b) Use (a) to define $\mathrm{tr}(T)$ for a linear transformation $T : V \to V$ where $\dim(V) < \infty$ (similarly to how we defined $\det(T)$ in Lecture 17). Your definition should not depend on the choice of a basis of V.

Problem 2: Let V be a vector space over a field F and $T : V \to V$ a linear map. Given a polynomial $p(x) \in \mathcal{P}(F)$, we can define $p(T)$ (the value of p at T) in a natural way: if $p(x) = \sum_{k=0}^{n} a_k x^k$ with $a_k \in F$, we set $p(T) = \sum_{k=0}^{n} a_k T^k$ (where $T^0 = I$, the identity map).

(a) Prove that $p(T)$ is a linear map and $\mathrm{Ker}\, p(T)$ and $\mathrm{Im}\, p(T)$ are both T-invariant subspaces of V.

(b) Let $v \in V$ be an eigenvector for T with eigenvalue λ. Prove that v is also an eigenvector for $p(T)$ with eigenvalue $p(\lambda)$.

The results of Problems 3 and 4 below will be essential for the proof of the existence of the Jordan Canonical Forms of matrices (tentatively scheduled for the Nov 9 week).

Problem 3: Let V be a finite-dimensional vector space over a field F, $n = \dim(V)$ and $T : V \to V$ a linear map. If W is a T-invariant subspace of V, denote by $T_W : W \to W$ the map obtained from T by restricting both the domain and codomain to W (we can restrict the codomain as well precisely because W is T-invariant).

Suppose now that $V = \bigoplus_{i=1}^{k} W_i$ where each V_i is T-invariant, and choose an ordered basis β_i for each V_i. Prove that

(i) $\beta = \beta_1 \cup \ldots \cup \beta_k$ is a basis of V. In (ii) below we view β as an ordered basis where elements of β_1 come first, followed by elements of β_2 etc.

(ii) The matrix $[T]_\beta$ is $k \times k$ block-diagonal, with i^{th} diagonal block equal to $[T_{W_i}]_{\beta_i}$, that is,

$$[T]_\beta = \begin{pmatrix} [T_{W_1}]_{\beta_1} & \ldots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \ldots & [T_{W_k}]_{\beta_k} \end{pmatrix}$$

Note: Part (i) has nothing to do with T. Also (i) holds without the assumption $\dim(V) < \infty$, but in the finite-dimensional case you can give a shorter proof using HW#8.6.

Problem 4: Let V be a vector space and $T : V \to V$ a linear map. In parts (b)–(e) assume that $\dim V < \infty$.

(a) Prove that $\text{Ker}(T^n) \subseteq \text{Ker}(T^{n+1})$ for each $n \geq 1$.

(b) By (a) we get an ascending chain of subspaces of V:

$$\text{Ker}(T) \subseteq \text{Ker}(T^2) \subseteq \ldots.$$

Prove that this chain stabilizes after finitely many steps, that is, there is $N \in \mathbb{N}$ s.t. $\text{Ker}(T^n) = \text{Ker}(T^N)$ for all $n \geq N$.

(c) Assume that $\text{Ker}(T) \cap \text{Im}(T) \neq \{0\}$. Prove that $\text{Ker}(T^2) \neq \text{Ker}(T)$.

(d) Use (b) and (c) (applied to a suitable power of T) to show that there exists $n \in \mathbb{N}$ s.t. $\text{Ker}(T^n) \cap \text{Im}(T^n) = \{0\}$. Deduce that $V = \text{Ker}(T^n) \oplus \text{Im}(T^n)$.

(e) Use (d) to show that if T is non-invertible and non-nilpotent, then V can be written as a direct sum of 2 nonzero T-invariant subspaces.

Problem 5: Let F be a field and $n \in \mathbb{N}$. Given an ordered basis β of F^n, let $Q(\beta)$ be the matrix whose columns are the elements of β (in the given order). Given $\lambda_1, \ldots, \lambda_n \in F$, let $D(\lambda_1, \ldots, \lambda_n)$ be the diagonal matrix with the diagonal entries $\lambda_1, \ldots, \lambda_n$ (in this order).

Prove that for a given $A \in \text{Mat}_n(F)$, an ordered basis $\beta = \{v_1, \ldots, v_n\}$ of F^n and scalars $\lambda_1, \ldots, \lambda_n \in F$, the following are equivalent:

(i) v_i is an eigenvector of A with eigenvalue λ_i for each $1 \leq i \leq n$;

(ii) $QD = AQ$ (or, equivalently, $Q^{-1}AQ = D$) where $Q = Q(\beta)$ and $D = D(\lambda_1, \ldots, \lambda_n)$.

Deduce from this that A is diagonalizable if and only if A is similar to a diagonal matrix.

Hint: Show that whenever the product of two matrices XY is defined, the jth column of XY is equal to $X\text{col}_j(Y)$ for all j. Note that the result of Problem 5 is stated in the book as Corollary after Theorem 5.1 on page 247, but as far as I can tell, the proof is not clearly presented.

Problem 6: In each of the following examples the field is $\mathbb{R}$ (real numbers). Find all eigenvalues and eigenvectors of the given $A \in \text{Mat}_3(\mathbb{R})$ and determine whether the matrix A is diagonalizable. If A is diagonalizable, find an invertible matrix Q and a diagonal matrix D s.t. $Q^{-1}AQ = D$.

$$\text{(a)} \quad A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 3 \end{pmatrix} \qquad \text{(b)} \quad A = \begin{pmatrix} 7 & -4 & 0 \\ 8 & -5 & 0 \\ 6 & -6 & 3 \end{pmatrix}$$

Problem 7: Let $A = \text{Mat}_n(F)$, and suppose that $\chi_T(x)$ splits completely over F, that is, $\chi_T(x) = \prod_{i=1}^{n}(x - \lambda_i)$ for some $\lambda_1, \ldots, \lambda_n \in F$. (Thus, $\lambda_1, \ldots, \lambda_n$ are precisely the eigenvalues of T listed with their algebraic multiplicities). Prove that

$$\sum_{i=1}^{n} \lambda_i = \text{tr}(A) \quad \text{and} \quad \prod_{i=1}^{n} \lambda_i = \det(A). \qquad (***)$$

Then explain why the analogous results holds if A is a replaced by a linear map $T : V \to V$ where V is a finite-dimensional vector space. **Hint:** Use HW#8.5.

Problem 8: Let V be a vector space, and let $T : V \to V$ and $S : V \to V$ be linear maps which commute ($TS = ST$). Let λ be an eigenvalue of T. Prove that the eigenspace $E_\lambda(T)$ is S-invariant.